

JUAN DELGADO

Data Science and Analytics Portfolio

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I. Executive Summary

Sales professional with 8 years in the optical industry, partnering with independent optometry offices to drive patient retention, profitability, and growth. With completing a Master of Science (M.S.) in Applied Data Science (AI Focus) at Syracuse University, I combine frontline commercial experience with machine learning to bridge the gap between sales intuition and analytic precision. **The projects that follow show how applied data science can turn insights into decisions a commercial team will actually act on.**

Professional Summary

Currently managing a \$52M portfolio at Kering Eyewear with 30% channel growth over three years and a consistent record of growth. Combines frontline commercial expertise with hands-on technical skills — Python, VBA, advanced Excel, SAP Business Objects and Salesforce — while completing an M.S. in Applied Data Science (AI Focus) at Syracuse University. Multilingual in Spanish, French, and German, with a track record of turning analysis into decisions that sales teams will execute to grow the business.

Key Accomplishments

- Grew \$52M strategic partnerships portfolio 30% over three years at Kering Eyewear, managing 20+ key accounts
 - Selected to manage two internal programs at Kering Eyewear: Field Sales Synergy Lead and Field Sales Internship
 - Built Python and VBA automation tools for sales reporting and lead generation, reducing manual workflows for both self and team
 - Exceeded new customer acquisition goal by 140% (28 of 20) in first year as Member Engagement Manager at CECOP USA
 - Delivered 99-105% of quota across four consecutive sales periods at CooperVision (FY18, FY19, Q4 FY20, FY21)
 - Recognized 4x on Commercial Highlights by President of Sales and named Sales Advisor at CooperVision (Oct 2019 – Dec 2021)
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II. Education

Master of Science in Applied Data Science (AI Focus) Syracuse University – Syracuse, NY 13244	May 2026
Bachelor of the Arts in International Business Bachelor of the Arts in French SUNY College at Brockport - Brockport, NY 14420	May 2016 May 2016

III. Skills

Languages

English (Native), Spanish (Native), German (Proficient), French (Proficient)

Microsoft Office

Excel, PowerPoint, Word, Outlook

Programming Languages

Python, R, VBA, SQL, HTML

Artificial Intelligence

Claude, ChatGPT, Gemini

Data Analysis & Visualization

Pandas, NumPy, Matplotlib, ggplot2, Data Cleaning, Data Wrangling, Machine Learning integration

Machine Learning

Supervised/Unsupervised Learning, Regression, Classification, Clustering, TensorFlow, Keras, PyTorch.

Other Relevant Skills

Public speaking, communication, presenting to stakeholders, statistical modeling, experimental design, problem-solving, teamwork.

IV. Projects

Project 1: Analyzing US Geographic Mobility and Median Income to Find Counties and States with Top Sales Growth Potential

IST 652 – Scripting for Data Analysis

Problem Statement

Businesses seeking to expand into new markets often rely on broad demographic assumptions or raw income data, which tend to favor large metropolitan areas and overlook emerging high-potential markets. This approach leads to inefficient territory planning and missed growth opportunities in smaller, rapidly developing regions.

This project addresses that gap by leveraging U.S. Census data on geographic mobility and median income to identify counties and states where rising income levels among newly arrived residents, combined with measurable population inflow, signal untapped sales growth potential. The goal is to produce targeted, data-driven recommendations that enable more precise and effective business development decisions.

Data Sources

This project utilized two primary data sources from the U.S. Census Bureau, selected for the reliability, geographic granularity, and relevance to measuring both income and population movement at the country level. Below are the datasets that were used to calculate and normalize migration inflow rates at the county level, validating that income trends were supported by actual residential migration patterns.

1. American Community Survey (ACS) — 1-Year Estimates

- **Provider:** U.S. Census Bureau
- **Years Used:** 2022 and 2023
- **Access Method:** Census Bureau API and downloadable tables

2. Current Population Survey (CPS) — Geographic Mobility Supplement

- **Provider:** U.S. Census Bureau
- **Years Used:** 2022 and 2023
- **Access Method:** Downloadable microdata files

Techniques & Tools

This project demonstrates a full data analysis pipeline, from raw data acquisition through final visualization, using industry-standard Python libraries and analytical methods. The following summarizes the core techniques and tools applied at each stage of the workflow.

Tool / Library	Purpose
Python	Primary programming language for all scripting and analysis
Pandas	Data cleaning, transformation, merging, and aggregation
NumPy	Numerical computations and array operations
GeoPandas	Geospatial data handling and county-level spatial joins
Contextily	Adding basemap tiles to geographic visualizations
Matplotlib	Chart creation and plot customization
U.S. Census API	Programmatic data retrieval from Census Bureau endpoints
Google Colaboratory	Interactive development, documentation, and result presentation

Results & Impact

The analysis produced clear, data-driven findings that moved beyond broad market assumptions to identify specific counties and regions with the strongest sales growth potential. Results were evaluated across two primary dimensions, income growth among newly arrived residents and population inflow rates, to highlight the most compelling business opportunities.

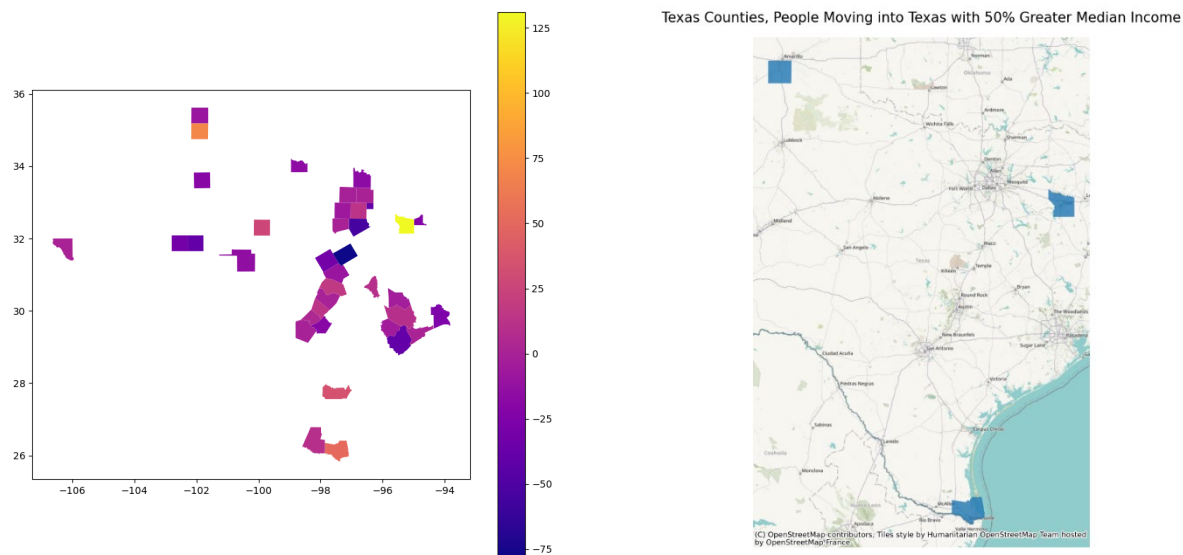
These counties showed the largest percentage increase in median income among newly arrived residents compared to the existing county baseline. They also highlight smaller, non-metropolitan counties that traditional raw income analysis would have overlooked entirely.

Rank	County	State	Income Increase
1	Geauga County	OH	340%
2	Saline County	AR	186%
3	Walton County	GA	177%

Visualizations

One core objective of this project was translating complex, multi-variable analysis into clear visual outputs that could communicate findings to both technical and non-technical audiences. All visualizations were produced using GeoPandas and Contextily in Python, with county-level census data joined to U.S. shapefile boundary files to enable geographic mapping.

The maps below show the top counties in Texas with potential growth opportunity.



Learnings & Next Steps

This project produced both technical and strategic insights that extended well beyond the immediate findings. By building a full data pipeline from raw census data to geographic visualizations and business recommendations, this project can be replicated in the future with updated census data. Furthermore, it would be interesting to analyze the changes of the top counties over time, as well as further research as to reasons why these dollars are flowing into specific counties for further business development strategy.

Link to Code

<https://github.com/juan-delgado329/us-census-data-analysis-for-geographic-sales-leads>

Project 2: Association Rule Mining B2B Eyewear Transactions

IST 718 – Big Data Analytics

Problem Statement

In large-scale B2B retail environments, sales teams and ordering platforms often lack the data-driven insight needed to identify which products are frequently purchased together. Without visibility into cross-product purchasing patterns, businesses miss opportunities to increase average order size, optimize product recommendations, and reinforce effective selling strategies.

This project addresses that gap by applying association rule mining to over 50,000 rows of B2B eyewear transaction data, using the FP-Growth algorithm in PySpark to uncover meaningful purchasing patterns across thousands of customer orders. The goal was to translate those patterns into actionable recommendations that can guide bundling strategies, inform field sales teams, and support product recommendation features within B2B ordering platforms.

Data Sources

This project utilized a single proprietary dataset drawn from real-world B2B eyewear transaction records. The dataset represented a large-scale, production-level source of sales data, making it well suited for applying big data analytics techniques and distributed computing tools.

Primary Dataset — B2B Eyewear Transaction Data

- **Source:** Internal B2B eyewear sales transaction records
- **Volume:** Approximately 10,000 transactions and over 50,000 rows of data
- **Granularity:** SKU-level transaction data capturing individual product purchases within each customer order

Key Characteristics

Attribute	Detail
Transaction Count	~10,000 orders
Row Count	50,000+ records
Product Coverage	50+ unique SKUs
Data Type	B2B wholesale purchase records
Notable Feature	Best-selling class designation by product

Techniques & Tools

This project applied big data analytics techniques to analyze large-scale B2B transaction data. The following summarizes the core methods and technologies used at each stage of the workflow, from data preparation through model tuning and business interpretation.

Tool / Library	Purpose
Python	Primary programming language for all scripting and analysis
PySpark	Distributed computing framework for processing large-scale transaction data
FP-Growth (MLlib)	Association rule mining algorithm for identifying frequent item sets
Pandas	Supporting data manipulation and preprocessing tasks
Google Colaboratory	Interactive development, documentation, and result presentation

Results & Impact

The analysis successfully identified meaningful purchasing patterns across thousands of B2B eyewear transactions, producing association rules and frequent item sets that translate directly into actionable sales and product recommendation strategies. Results were evaluated across three iterative dataset configurations, with each iteration revealing progressively deeper insights into cross-product purchasing behavior.

The FP-Growth algorithm identified frequent item sets appearing consistently across a significant number of transactions, confirming that the purchasing patterns discovered were repeatable behaviors rather than isolated occurrences.

Metric	Result
Transaction Range of Item Sets	134 to 497 transactions
Total Transactions Analyzed	~10,000
Total Rows Processed	50,000+
Top 30 Item Sets Containing KPS Products	23 out of 30

The presence of Best-Selling products in 23 of the top 30 frequent item sets was a particularly significant finding. Although Best-Selling products represented only about one-fourth of the total SKUs, their dominance in the results validated that the existing sales strategy around these products is working, and provided data driven evidence to support continued investment on Best-Selling product focused initiatives.

Learnings & Next Steps

This project demonstrated that large scale transaction data contains rich, actionable intelligence that can directly improve sales performance, inform product strategy, and strengthen customer relationships. Analyzing future data sets would allow us to evaluate Best-Selling product initiatives with the goal of implementing a recommendation system in our online purchasing platform.

Link to Code

<https://github.com/juan-delgado329/association-rule-mining-eyewear-transactions>

Project 3: Convolutional Neural Networks using iPhone Images for Eyewear Classification

IST 691 – Deep Learning in Practice

Problem Statement

Sales teams operating in field environments often struggle to quickly and accurately identify eyewear products, particularly when working across large catalogs of visually similar styles and brands. Without a reliable, automated way to classify products, teams must rely on manual lookups or prior product knowledge, creating inefficiencies that slow down the sales process and increase the risk of misidentification.

This project addresses that gap by building and evaluating convolutional neural network (CNN) models capable of classifying eyewear products directly from iPhone images captured in realistic uncontrolled sales environments. The goal is to determine whether deep learning techniques, including transfer learning and hierarchical classification, can achieve strong classification accuracy. With the goal to support practical business applications such as automatic product identification, inventory tracking, and mobile based recommendation tools that can help sales teams operate more efficiently and drive higher value customer interactions.

Data Sources

This project utilized a proprietary dataset of eyewear product images captured specifically for this analysis. Unlike many deep learning projects that rely on large publicly available image datasets, this project was designed to reflect the realities of a field sales environment, where images are taken under imperfect, real-world conditions.

Primary Dataset — iPhone Eyewear Product Images

- **Source:** Original images captured using an iPhone camera
- **Subject Matter:** Eyewear products across multiple brands, styles, and product types
- **Access Method:** Directly captured and compiled for this project
- **Environment:** Realistic, uncontrolled sales conditions rather than professionally lit or staged photography

Key Characteristics

Attribute	Detail
Image Source	iPhone camera captures
Classification Levels	Fine level (individual style), product type, and brand
Environment	Uncontrolled, real-world sales conditions
Dataset Size	Limited dataset typical of a real-world business setting
Product Coverage	Multiple eyewear brands and styles across several product categories

Techniques & Tools

This project applied advanced deep learning techniques to evaluate multiple convolutional neural network architectures for eyewear product classification. The following summarizes the core techniques and tools used throughout the modeling, evaluation, and comparison process.

Tool / Library	Purpose
Python	Primary programming language for all modeling and analysis
TensorFlow / Keras	Deep learning framework for building, training, and evaluating CNN models
MobileNetV2	Lightweight pretrained model used for transfer learning experiments
VGG16	Pretrained model used for hierarchical CNN transfer learning
VGG19	Deeper pretrained model evaluated as a comparison to VGG16
NumPy	Numerical computations and array operations
Matplotlib	Visualization of model training history, accuracy, and loss curves
Google Colaboratory	Interactive Python development, documentation, and result presentation

Results & Impact

The analysis successfully evaluated multiple convolutional neural network architectures for eyewear product classification, progressing from a poorly performing baseline model to a highly accurate transfer learning solution capable of supporting real-world business applications. Results were measured consistently across accuracy and loss metrics, with each modeling iteration building on the lessons of the previous one to produce progressively stronger outcomes.

The project systematically evaluated six model configurations to identify the most effective approach for classifying luxury eyewear products from real-world iPhone images. As the comparison table shows, the gap between scratch-trained and transfer learning models was stark: the Baseline CNN and Hierarchical CNN without transfer learning both failed to exceed 15% validation accuracy, confirming that pretrained features — not architecture — are the deciding factor when data is limited. Transfer learning with MobileNetV2 pushed accuracy to 89.7%, and fine-tuning that same model brought validation loss down from 0.5371 to 0.4684, a meaningful gain in prediction confidence even as accuracy remained comparable. The Hierarchical CNN with VGG16 delivered the most practically valuable result at 84.5% fine-level accuracy, and its strong performance across brand and product-type levels simultaneously makes it the strongest candidate for field sales deployment. The VGG16 vs. VGG19 comparison rounds out the table with an important caveat: the deeper model underperformed on both accuracy and consistency, reinforcing that with constrained datasets, added complexity is a liability rather than an advantage.

Full Model Performance Comparison

Model	Validation Accuracy	Validation Loss	Key Takeaway
Baseline CNN	14.4%	—	Scratch training ineffective with limited data
MobileNetV2 Transfer Learning	89.7%	0.5371	Transfer learning dramatically improves performance
MobileNetV2 Fine-Tuned	88.2%	0.4684	Fine-tuning improves prediction confidence
Hierarchical CNN (No Transfer)	13.1%	—	Architecture alone insufficient without pretrained features
Hierarchical CNN + VGG16	84.5%	—	Best approach for visually similar multi-level classification
VGG19 Comparison	Below VGG16	Less consistent	Deeper architecture did not improve results

Learnings & Next Steps

This project demonstrated that advanced deep learning techniques can be applied effectively to real-world business problems even under realistic constraints like limited data and uncontrolled image conditions. Deploying this project would lead to a practical and deployable capability that can directly drive revenue growth, improve sales productivity, and create competitive advantage in a field sales environment.

Link to Code

<https://github.com/juan-delgado329/cnn-iphone-images-eyewear-classification>

V. Additional Recognition**Presentations**

25 Presentations on Luxury Eyewear, the Optical Industry and Profitability for Kering Eyewear **2023 – Present**

- *In person audiences ranged from 10 to 100 people*

Presented on sales article research at CooperVision Sales Conference

December 2019

Awards & Achievements

Manager of Synergy Lead Program at Kering Eyewear

January 2026 – Present

Manager of Internship Program

April 2025 – Present

Company Values Award at CECOP USA

Q1 and Q3 2022

Sales Advisor at CooperVision

October 2019 – December 2021

Featured 4 times, Commercial Highlights by President of Sales

Sept 2018, Nov 2020, Jan 2021, Mar 2021